

Projected Gradient Descent: Algorithm and Convergence Analysis

1 Algorithm Description

The algorithm iteratively updates parameters by taking gradient steps and projecting the result back onto the constraint set at each iteration. It is designed for smooth convex optimization problems with closed, convex, and bounded constraint domains.

2 Pseudo Code and Dry Run

Algorithm Name

Projected Gradient Descent (PGD)

Mathematical Formulation

Problem Definition: Minimize a smooth convex objective function $f(\mathbf{w})$ subject to $\mathbf{w} \in \mathcal{C}$, where: - $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable with Lipschitz continuous gradient. - $\mathcal{C} \subseteq \mathbb{R}^n$ is a closed, convex, and bounded constraint set.

Update Rule: At iteration t , compute:

$$\mathbf{w}_{t+1} = \Pi_{\mathcal{C}}(\mathbf{w}_t - \eta_t \nabla f(\mathbf{w}_t)),$$

where: - $\nabla f(\mathbf{w}_t)$ is the gradient of f at $\mathbf{w}_t$. - $\eta_t > 0$ is the step size (learning rate) at iteration t . - $\Pi_{\mathcal{C}}(\mathbf{v}) = \arg \min_{\mathbf{w} \in \mathcal{C}} \|\mathbf{w} - \mathbf{v}\|_2^2$ is the Euclidean projection onto $\mathcal{C}$.

Hyperparameters: 1. **Step size** η_t : Controls the magnitude of the gradient step. Common choices include constant $\eta_t = \eta$ or diminishing schedules (e.g., $\eta_t = O(1/\sqrt{t})$). 2. **Initialization:** $\mathbf{w}_0 \in \mathcal{C}$. 3. **Termination Conditions:** Maximum number of iterations T , or convergence tolerance ϵ for $\|\nabla f(\mathbf{w}_t)\|$.

Pseudocode

Input: Objective function f , constraint set $\mathcal{C}$, step size η , initialization $\mathbf{w}_0 \in \mathcal{C}$, maximum iterations T , tolerance ϵ . **Output:** Approximate solution $\mathbf{w}_T$. $t = 0, 1, \dots, T - 1$ Compute gradient: $\mathbf{g}_t \leftarrow \nabla f(\mathbf{w}_t)$. Gradient step: $\tilde{\mathbf{w}}_{t+1} \leftarrow \mathbf{w}_t - \eta \cdot \mathbf{g}_t$. Projection: $\mathbf{w}_{t+1} \leftarrow \Pi_{\mathcal{C}}(\tilde{\mathbf{w}}_{t+1})$. Check stopping condition: $\|\mathbf{g}_t\| < \epsilon \Rightarrow \text{break}$. **Return** $\mathbf{w}_{t+1}$.

Dry Run Example

Objective Function: $f(w) = (w - 3)^2$, constraint set $\mathcal{C} = [0, 5]$. **Initialization:** $w_0 = 0$, step size $\eta = 0.1$, deterministic gradient. **Projection Rule:** For $\tilde{w} \notin \mathcal{C}$, project to the nearest endpoint (i.e., $\Pi_{\mathcal{C}}(\tilde{w}) = \max(0, \min(5, \tilde{w}))$).

Iteration 1: 1. **Gradient:** $\nabla f(w_0) = 2(w_0 - 3) = 2(0 - 3) = -6$. 2. **Gradient Step:** $\tilde{w}_1 = 0 - 0.1 \cdot (-6) = 0.6$. 3. **Projection:** $0.6 \in [0, 5] \Rightarrow w_1 = 0.6$. 4. **Objective Value:** $f(w_1) = (0.6 - 3)^2 = 5.76$.

Iteration 2: 1. **Gradient:** $\nabla f(w_1) = 2(0.6 - 3) = -4.8$. 2. **Gradient Step:** $\tilde{w}_2 = 0.6 - 0.1 \cdot (-4.8) = 1.08$. 3. **Projection:** $1.08 \in [0, 5] \Rightarrow w_2 = 1.08$. 4. **Objective Value:** $f(w_2) = (1.08 - 3)^2 = 3.6864$.

Iteration 3: 1. **Gradient:** $\nabla f(w_2) = 2(1.08 - 3) = -3.84$. 2. **Gradient Step:** $\tilde{w}_3 = 1.08 - 0.1 \cdot (-3.84) = 1.464$. 3. **Projection:** $1.464 \in [0, 5] \Rightarrow w_3 = 1.464$. 4. **Objective Value:** $f(w_3) = (1.464 - 3)^2 = 2.3593$.

Summary: The algorithm iteratively reduces the objective value by descending along the gradient and projecting onto $\mathcal{C}$. In this example, the constraint set $\mathcal{C} = [0, 5]$ does not actively restrict updates, but the projection step remains formally correct.

3 Convergence Theory

Assumptions

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth convex objective function and $\mathcal{K} \subset \mathbb{R}^n$ a non-empty, closed, convex, and bounded constraint set. We impose the following assumptions:

1. **Smoothness:** The gradient ∇f is L -Lipschitz continuous, i.e., for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|.$$

2. **Convexity:** The function f is convex:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle, \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{K}.$$

3. **Constraint Set:** The constraint set $\mathcal{K}$ has diameter $D < \infty$, where

$$D = \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{K}} \|\mathbf{x} - \mathbf{y}\|.$$

4. **Gradient Boundedness:** There exists $G < \infty$ such that $\|\nabla f(\mathbf{x})\| \leq G$ for all $\mathbf{x} \in \mathcal{K}$.

Main Convergence Theorem

[Convergence Rate] Let $\{\mathbf{x}_t\}_{t=0}^T$ be generated by the projected gradient descent update:

$$\mathbf{x}_{t+1} = \text{Proj}_{\mathcal{K}}(\mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)),$$

with step size $\eta = 1/L$. Then, the average iterate $\bar{\mathbf{x}}_T = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t$ satisfies:

$$f(\bar{\mathbf{x}}_T) - f^* \leq \frac{LD^2}{2T},$$

where $f^* = \min_{\mathbf{x} \in \mathcal{K}} f(\mathbf{x})$. Thus, the algorithm achieves an $\mathcal{O}(1/T)$ convergence rate.

Using the L -smoothness of f , we derive:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \frac{1}{2L} \|\nabla f(\mathbf{x}_t)\|^2.$$

By convexity and projection properties, we obtain:

$$2(f(\mathbf{x}_t) - f^*) \leq L \left(\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 + \frac{1}{L^2} \|\nabla f(\mathbf{x}_t)\|^2 \right).$$

Summing over $t = 1$ to T , telescoping bounds yield:

$$\sum_{t=1}^T (f(\mathbf{x}_t) - f^*) \leq \frac{LD^2}{2}.$$

The result follows by averaging both sides.

Theoretical Properties

1. **Stability:** All iterates $\mathbf{x}_t \in \mathcal{K}$ due to the projection operator, ensuring boundedness by the constraint set's diameter D .
2. **Hyperparameter Sensitivity:** The step size $\eta = 1/L$ is optimal. Larger η may destabilize convergence, while smaller η reduces progress per iteration.
3. **Comparison to Baselines:**
 - *Unconstrained Gradient Descent:* Matches the $\mathcal{O}(1/T)$ rate for smooth functions but extends to constrained domains.
 - *Frank-Wolfe:* Avoids projections but may require line searches and often has slower convergence for non-linear constraints.
4. **Special Cases:**
 - *Strong Convexity:* If f is μ -strongly convex, the rate improves to $\mathcal{O}(e^{-\mu T/L})$ (linear convergence).
 - *Unconstrained Solution:* If $\mathbf{x}^* \in \text{int}(\mathcal{K})$, the method reduces to standard gradient descent.

Discussion

This analysis establishes that projected gradient descent achieves an $\mathcal{O}(1/T)$ convergence rate for smooth convex optimization over closed, convex, bounded domains. The projection operation ensures feasibility without compromising the convergence rate achievable in unconstrained settings. Critical dependencies include the Lipschitz constant L and domain diameter D , which scale the error bound linearly and quadratically, respectively. The requirement to know L for step-size selection highlights a limitation, suggesting potential improvements via adaptive or line-search methods. Compared to Frank-Wolfe, the algorithm excels in settings where projections onto $\mathcal{K}$ are computationally tractable. Future work could extend these results to stochastic or non-convex regimes.

4 Convergence Proofs

Projected Gradient Descent: Convergence Analysis

Preliminaries (Definitions and Lemmas)

Definitions **Definition 1 (Smoothness):** A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth if its gradient is L -Lipschitz continuous:

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

Definition 2 (Convexity): A function f is convex if for all $\mathbf{x}, \mathbf{y} \in \mathcal{K}$ and $\lambda \in [0, 1]$:

$$f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}).$$

Definition 3 (Projection Operator): For a closed convex set $\mathcal{K} \subset \mathbb{R}^n$, the projection of $\mathbf{z} \in \mathbb{R}^n$ onto $\mathcal{K}$ is:

$$\text{Proj}_{\mathcal{K}}(\mathbf{z}) = \arg \min_{\mathbf{x} \in \mathcal{K}} \|\mathbf{x} - \mathbf{z}\|^2.$$

Lemmas **Lemma 1 (Descent Lemma):** For L -smooth f :

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2 \quad \forall \mathbf{x}, \mathbf{y}.$$

Lemma 2 (Convexity Inequality): For convex f :

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{K}.$$

Lemma 3 (Projection Theorem): For any $\mathbf{z} \in \mathbb{R}^n$ and $\mathbf{x} \in \mathcal{K}$:

$$\|\text{Proj}_{\mathcal{K}}(\mathbf{z}) - \mathbf{x}\|^2 \leq \|\mathbf{z} - \mathbf{x}\|^2 - \|\text{Proj}_{\mathcal{K}}(\mathbf{z}) - \mathbf{z}\|^2.$$

Lemma 4 (Pythagorean Theorem for Projections): For all $\mathbf{x} \in \mathcal{K}$:

$$\langle \mathbf{z} - \text{Proj}_{\mathcal{K}}(\mathbf{z}), \mathbf{x} - \text{Proj}_{\mathcal{K}}(\mathbf{z}) \rangle \leq 0.$$

Main Proof (Step-by-Step Derivation)

Algorithm Description Projected Gradient Descent (PGD) updates:

$$\mathbf{x}_{t+1} = \text{Proj}_{\mathcal{K}} \left(\mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t) \right),$$

with step size $\eta = \frac{1}{L}$.

Key Inequality Derivation Let $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{K}} f(\mathbf{x})$. Apply Lemma 3 with $\mathbf{z} = \mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t)$ and $\mathbf{x} = \mathbf{x}^*$:

$$\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \leq \left\| \mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t) - \mathbf{x}^* \right\|^2 - \left\| \mathbf{x}_{t+1} - \left(\mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t) \right) \right\|^2.$$

Expand the first squared norm:

$$\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \frac{2}{L} \langle \nabla f(\mathbf{x}_t), \mathbf{x}_t - \mathbf{x}^* \rangle + \frac{1}{L^2} \|\nabla f(\mathbf{x}_t)\|^2.$$

Substitute into the inequality:

$$\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \frac{2}{L} \langle \nabla f(\mathbf{x}_t), \mathbf{x}_t - \mathbf{x}^* \rangle + \frac{1}{L^2} \|\nabla f(\mathbf{x}_t)\|^2.$$

Apply Lemma 2 (convexity):

$$\langle \nabla f(\mathbf{x}_t), \mathbf{x}_t - \mathbf{x}^* \rangle \geq f(\mathbf{x}_t) - f^*.$$

Substitute this into the inequality:

$$\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \frac{2}{L} (f(\mathbf{x}_t) - f^*) + \frac{1}{L^2} \|\nabla f(\mathbf{x}_t)\|^2.$$

Telescoping Sum Rearrange terms:

$$\frac{2}{L} (f(\mathbf{x}_t) - f^*) \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 + \frac{1}{L^2} \|\nabla f(\mathbf{x}_t)\|^2.$$

Sum over $t = 1$ to T :

$$\frac{2}{L} \sum_{t=1}^T (f(\mathbf{x}_t) - f^*) \leq \|\mathbf{x}_1 - \mathbf{x}^*\|^2 - \|\mathbf{x}_{T+1} - \mathbf{x}^*\|^2 + \frac{1}{L^2} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|^2.$$

Since $\|\mathbf{x}_{T+1} - \mathbf{x}^*\|^2 \geq 0$ and $\|\mathbf{x}_1 - \mathbf{x}^*\| \leq D$:

$$\sum_{t=1}^T (f(\mathbf{x}_t) - f^*) \leq \frac{LD^2}{2} + \frac{1}{2L} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|^2.$$

Bounding Gradient Norms via Smoothness Apply Lemma 1 (Descent Lemma) with $\mathbf{y} = \mathbf{x}_{t+1}$ and $\mathbf{x} = \mathbf{x}_t$:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.$$

From the projection update $\mathbf{x}_{t+1} = \text{Proj}_{\mathcal{K}}(\mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t))$, Lemma 4 implies:

$$\langle \mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t) - \mathbf{x}_{t+1}, \mathbf{x}_{t+1} - \mathbf{x}_t \rangle \geq 0.$$

Expand this:

$$-\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 - \frac{1}{L} \langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle \geq 0.$$

Rearrange:

$$\langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle \leq -L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.$$

Substitute into the Descent Lemma inequality:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - L\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 + \frac{L}{2}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 = f(\mathbf{x}_t) - \frac{L}{2}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.$$

This implies:

$$\frac{1}{L^2}\|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{2}{L}(f(\mathbf{x}_t) - f(\mathbf{x}_{t+1})).$$

Sum over $t = 1$ to T :

$$\frac{1}{L^2} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{2}{L}(f(\mathbf{x}_1) - f(\mathbf{x}_{T+1})) \leq \frac{2}{L}(f(\mathbf{x}_1) - f^*).$$

Substitute into the earlier bound:

$$\sum_{t=1}^T (f(\mathbf{x}_t) - f^*) \leq \frac{LD^2}{2} + \frac{1}{2L} \cdot 2L(f(\mathbf{x}_1) - f^*) = \frac{LD^2}{2} + (f(\mathbf{x}_1) - f^*).$$

However, this introduces an additional constant term. To eliminate this, revisit the key inequality derived from combining smoothness and projection:

$$\frac{2}{L}(f(\mathbf{x}_t) - f^*) \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 + \frac{2}{L}(f(\mathbf{x}_t) - f(\mathbf{x}_{t+1})).$$

Cancel $\frac{2}{L}f(\mathbf{x}_t)$ on both sides:

$$\frac{2}{L}f^* \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 - \frac{2}{L}f(\mathbf{x}_{t+1}).$$

Rearrange:

$$\frac{2}{L}f(\mathbf{x}_{t+1}) \leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 + \frac{2}{L}f^*.$$

Sum over $t = 1$ to T :

$$\frac{2}{L} \sum_{t=2}^{T+1} f(\mathbf{x}_t) \leq \|\mathbf{x}_1 - \mathbf{x}^*\|^2 - \|\mathbf{x}_{T+1} - \mathbf{x}^*\|^2 + \frac{2}{L}Tf^*.$$

Drop the negative term $-\|\mathbf{x}_{T+1} - \mathbf{x}^*\|^2$:

$$\frac{2}{L} \sum_{t=2}^{T+1} f(\mathbf{x}_t) \leq \frac{2}{L}Tf^* + D^2.$$

Adjust indices to sum from $t = 1$ to T :

$$\frac{2}{L} \sum_{t=1}^T f(\mathbf{x}_t) \leq \frac{2}{L}Tf^* + D^2.$$

Divide both sides by T :

$$\frac{2}{L} \cdot \frac{1}{T} \sum_{t=1}^T f(\mathbf{x}_t) \leq \frac{2}{L}f^* + \frac{D^2}{T}.$$

Subtract $\frac{2}{L}f^*$:

$$\frac{2}{L} \left(\frac{1}{T} \sum_{t=1}^T f(\mathbf{x}_t) - f^* \right) \leq \frac{D^2}{T}.$$

Multiply both sides by $\frac{L}{2}$:

$$\frac{1}{T} \sum_{t=1}^T f(\mathbf{x}_t) - f^* \leq \frac{LD^2}{2T}.$$

By convexity:

$$f\left(\frac{1}{T} \sum_{t=1}^T \mathbf{x}_t\right) \leq \frac{1}{T} \sum_{t=1}^T f(\mathbf{x}_t).$$

Thus:

$$f(\bar{\mathbf{x}}_T) - f^* \leq \frac{LD^2}{2T}.$$

Rate Derivation (With Constants)

The derived bound explicitly includes constants:

$$f(\bar{\mathbf{x}}_T) - f^* \leq \frac{LD^2}{2T}.$$

This shows that the suboptimality gap decreases at a rate of $\mathcal{O}(1/T)$, where L is the smoothness constant and D is the diameter of the constraint set $\mathcal{K}$.

Special Cases

Case 1: Strong Convexity If f is μ -strongly convex (i.e., $f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2$), the convergence rate improves to linear. The key inequality becomes:

$$\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \leq \left(1 - \frac{\mu}{L}\right) \|\mathbf{x}_t - \mathbf{x}^*\|^2,$$

leading to:

$$f(\bar{\mathbf{x}}_T) - f^* \leq \frac{LD^2}{2} e^{-\mu T/L}.$$

Case 2: Unconstrained Solution If $\mathbf{x}^* \in \text{int}(\mathcal{K})$, the projection operation has no effect in a neighborhood of $\mathbf{x}^*$. The iterates reduce to standard gradient descent:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \frac{1}{L} \nabla f(\mathbf{x}_t),$$

with the same $\mathcal{O}(1/T)$ rate for smooth convex functions.

Conclusion

Projected Gradient Descent achieves an $\mathcal{O}(1/T)$ convergence rate for smooth convex optimization over closed, convex, bounded domains. The analysis leverages smoothness, convexity, and projection properties to derive tight bounds on the suboptimality gap. Special cases like strong convexity and unconstrained solutions further refine the convergence behavior.